EEES302 - Digital Design

Experiment 5

Decoders and Multiplexers

1) Objective:

The objective of this laboratory is to investigate the use of decoders and multiplexers to implement combinational logic circuits.

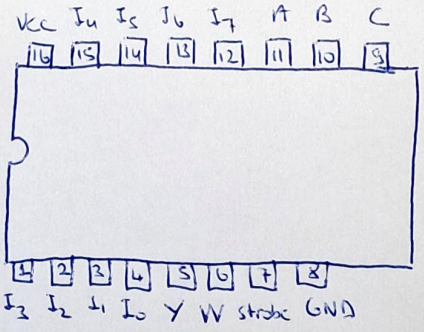
2) Review of Experiment:

Decoders:
A decoder is a combinational logic circuit that activates a specific output line based on an input binary or BCD code.

Multiplexers:
A multiplexer (MUX), or data selector, routes one of several data inputs to a single output based on select lines and can be highly efficient for implementing combinational logic. A multiplexer implementation table is used to systematically determine the correct connections for the data inputs.

3) Equipment List:

The experiment requires 8x1 MUX IC.



74151 8x1 MUX IC

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4) Preliminary Work:

Circuit 1:

	$BC'=00$	$BC'=01$	$BC'=10$	$BC=11$
$\bar{A}$	(0)	1	2	(3)
A	(4)	(5)	6	7
	1	A	0	$\bar{A}$

$F(A,B,C) = \Sigma m(0,3,4,5)$

Circuit 2:

	$CB'=00$	$CB=01$	$CB'=10$	$CB=11$
$\bar{A}$	(0)	(2)	1	(3)
A	(4)	(6)	5	7
	1	A	0	$\bar{A}$

$F(A,B,C) = \Sigma m(0,3,4,6)$

Circuit 3:

	$AC'=00$	$A'C=01$	$AC'=10$	$AC=11$
$\bar{B}$	(0)	1	4	(5)
B	(2)	(3)	6	7
	1	B	0	$\bar{B}$

$F(A,B,C) = \Sigma m(0,2,3,5)$

Circuit 4:

	$CA'=00$	$CA=01$	$CA'=10$	$CA=11$
$\bar{B}$	(0)	4	1	(5)
B	(2)	(6)	3	7
	1	B	0	$\bar{B}$

$F(A,B,C) = \Sigma m(0,2,5,6)$

Circuit 5:

	$A'B=00$	$A'B=01$	$AB'=10$	$AB=11$
$\bar{C}$	(0)	2	4	(6)
C	(1)	(3)	5	7
	1	C	0	$\bar{C}$

$F(A,B,C) = \Sigma m(0,1,3,6)$

Circuit 6:

	$B'A=00$	$B'A=01$	$BA'=10$	$BA=11$
$\bar{C}$	(0)	4	2	(6)
C	(1)	(5)	3	7
	1	C	0	$\bar{C}$

$F(A,B,C) = \Sigma m(0,1,5,6)$

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Decoders and Encoders

5) Experimental Procedure:

Implament $F(A,B,C,D) = \Sigma(1,3,5,6,9,14,15)$ function with 8-to-1 data selector/multiplexer as shown.

A	I_0	I_1	I_2	I_3	I_4	I_5	I_6	I_7
$\bar{A}$	0	(1)	2	(3)	4	(5)	(6)	7
A	8	(9)	10	11	12	13	(14)	(15)
	0	1	0	$\bar{A}$	0	$\bar{A}$	1	A

The multiplexers input pins are connected as found in multiplexer implementation table shown above.

The output is tested by changing the binary order of select lines.

The circuit is built on breadboard with jumpers and tested with ossilator.

6) Experimental Results:

Upon constructing the circuit using 74151 8x1 multiplexer and setting the input variables (A,B,C,D) via jumpers, the output Y exactly mimics the designed logic function.

This experiment confirms that complex combinational logic equations can be robustly simplified and implamented on hardware strictly using a single multiplexer and an inverter, heavily reducing the total gate count compared to standard AND-OR logic networks.

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